PROBLEM9:

PROBLEM STATEMENT:

Multi-Product Shopping Budget Constraint: Evaluate a shopping basket containing multiple products and quantities, apply discount conditions, and determine which purchases must be reduced to remain within a fixed budget.

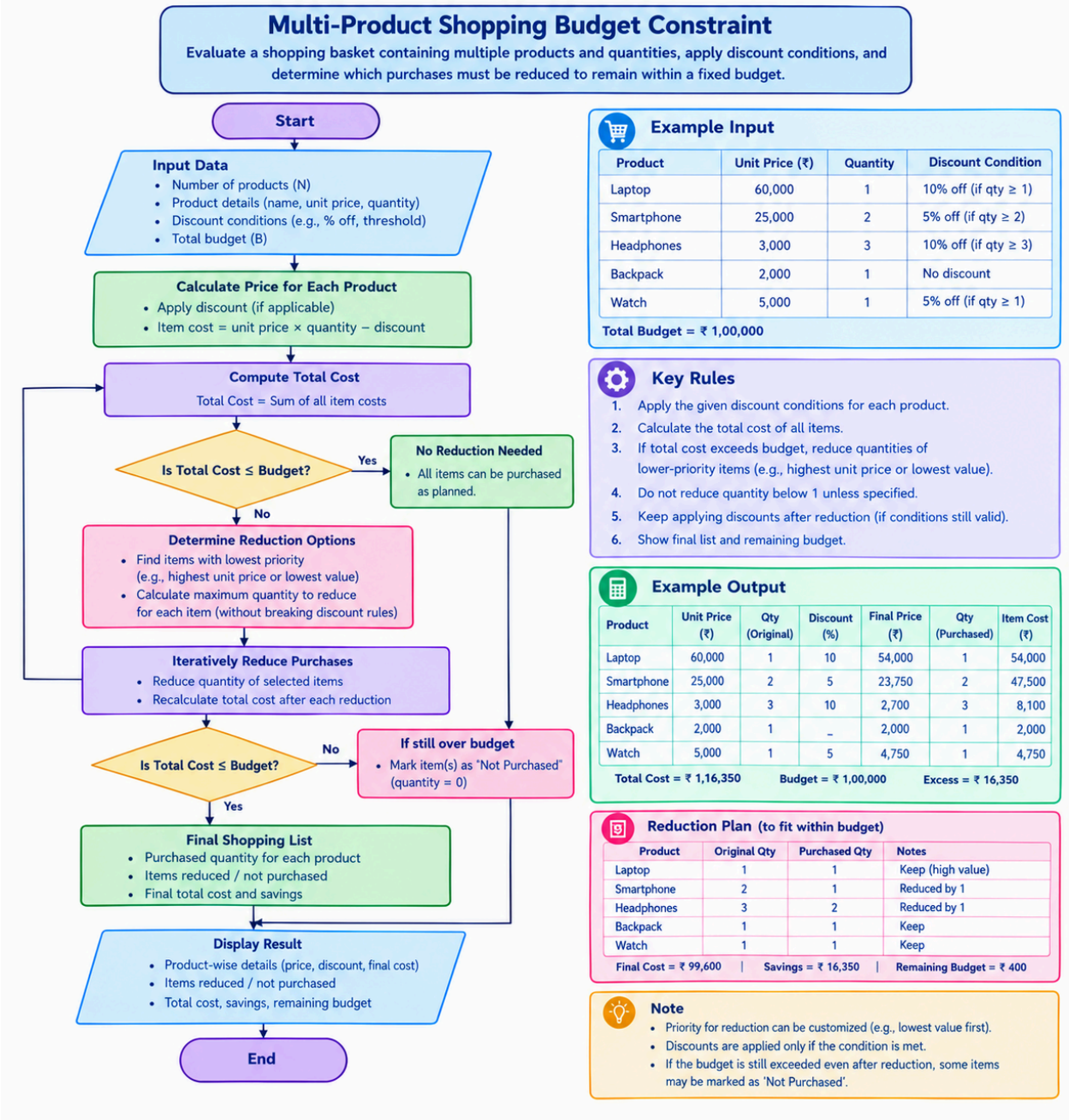
Objective

To develop a C program that evaluates a shopping basket containing multiple products and quantities, calculates the total cost, applies discount conditions, compares the final amount with a fixed budget, and identifies which product quantities must be reduced to stay within the budget.

2. Algorithm

1. Start the program.
2. Input the shopping budget and the number of products.
3. For each product, input its name, unit price, and quantity.
4. Validate the budget, prices, and quantities.
5. Calculate the cost of each product by multiplying its unit price by its quantity.
6. Calculate the subtotal of all products.
7. Apply a 10% discount if the subtotal is ₹5,000 or more.
8. Calculate the final amount after discount.
9. Compare the final amount with the fixed budget.
10. If the amount exceeds the budget, identify products whose quantities can be reduced.
11. Reduce quantities, starting with the most expensive products, until the budget is satisfied.
12. Display the original subtotal, discount, final amount, reduced quantities, and remaining budget.
13. Stop the program.

FLOWCHART:



PROGRAM:

```
#include <stdio.h>

int main() {
```

```
int n, i, j, tempQty;
char name[20][30], tempName[30];
int qty[20];
float price[20], tempPrice;
float budget, subtotal = 0, discount, total;
float originalSubtotal, originalDiscount;
int reduced = 0;

printf("Enter your shopping budget (Rs.): ");
scanf("%f", &budget);

printf("Enter number of products (1-20): ");
scanf("%d", &n);

if (budget < 0 || n < 1 || n > 20) {
    printf("Invalid budget or number of products!\n");
    return 0;
}

for (i = 0; i < n; i++) {
    printf("\nEnter product name: ");
    scanf("%29s", name[i]);

    printf("Enter unit price: ");
    scanf("%f", &price[i]);

    printf("Enter quantity: ");
    scanf("%d", &qty[i]);

    if (price[i] < 0 || qty[i] < 0) {
        printf("Invalid price or quantity!\n");
        return 0;
    }

    subtotal += price[i] * qty[i];
}

originalSubtotal = subtotal;
originalDiscount = (subtotal >= 5000) ?
                subtotal * 0.10f : 0;
total = subtotal - originalDiscount;

printf("\n--- ORIGINAL SHOPPING BILL ---\n");
printf("Subtotal: Rs.%.2f\n", originalSubtotal);
```

```
printf("Discount: Rs.%.2f\n", originalDiscount);
printf("Final Amount: Rs.%.2f\n", total);
printf("Budget: Rs.%.2f\n", budget);

if (total <= budget) {
    printf("Shopping is within budget.\n");
} else {
    printf("\nBudget exceeded by Rs.%.2f\n",
           total - budget);

    /* Sort products by unit price, highest first */
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (price[j] > price[i]) {
                tempPrice = price[i];
                price[i] = price[j];
                price[j] = tempPrice;

                tempQty = qty[i];
                qty[i] = qty[j];
                qty[j] = tempQty;

                /* Swap product names */
                {
                    char temp[30];
                    int k;
                    for (k = 0; k < 30; k++) {
                        temp[k] = name[i][k];
                        name[i][k] = name[j][k];
                        name[j][k] = temp[k];
                    }
                }
            }
        }
    }

    /* Reduce expensive products first */
    for (i = 0; i < n && total > budget; i++) {
        while (qty[i] > 0 && total > budget) {
            qty[i]--;
            reduced++;

            subtotal = 0;
            for (j = 0; j < n; j++)
```

```
                subtotal += price[j] * qty[j];

            discount = (subtotal >= 5000) ?
                    subtotal * 0.10f : 0;
            total = subtotal - discount;
        }
    }

    printf("\n--- REVISED SHOPPING BASKET ---\n");
    for (i = 0; i < n; i++) {
        printf("%s: Quantity = %d, Cost = Rs.%.2f\n",
            name[i], qty[i], price[i] * qty[i]);
    }

    discount = (subtotal >= 5000) ?
            subtotal * 0.10f : 0;
    total = subtotal - discount;

    printf("Revised Subtotal: Rs.%.2f\n", subtotal);
    printf("Discount: Rs.%.2f\n", discount);
    printf("Final Amount: Rs.%.2f\n", total);
    printf("Budget Remaining: Rs.%.2f\n",
        budget - total);
    printf("Units reduced: %d\n", reduced);
  }

  return 0;
}
```

PSEUDOCODE:

```
BEGIN

INPUT budget
INPUT number_of_products

IF budget < 0 OR number_of_products is invalid
  PRINT "Invalid input"
  STOP
END IF

FOR each product
  INPUT product_name
```

```
    INPUT unit_price
    INPUT quantity

    IF unit_price < 0 OR quantity < 0
        PRINT "Invalid product details"
        STOP
    END IF

    product_cost = unit_price * quantity
    subtotal = subtotal + product_cost
END FOR

IF subtotal >= 5000
    discount = subtotal * 10 / 100
ELSE
    discount = 0
END IF

total = subtotal - discount

IF total <= budget
    PRINT "Shopping is within budget"
ELSE
    SORT products by unit price in descending order

    FOR each product
        WHILE quantity > 0 AND total > budget
            REDUCE quantity by 1
            RECALCULATE subtotal

            IF subtotal >= 5000
                discount = subtotal * 10 / 100
            ELSE
                discount = 0
            END IF

            total = subtotal - discount
        END WHILE
    END FOR
END IF

DISPLAY revised product quantities
DISPLAY subtotal, discount and final amount
DISPLAY remaining budget
```

END

OUTPUT:

Example: Shopping budget ₹5,000

Enter your shopping budget (Rs.): 5000
Enter number of products (1-20): 3

Enter product name: Rice
Enter unit price: 1200
Enter quantity: 2

Enter product name: Oil
Enter unit price: 800
Enter quantity: 2

Enter product name: Shoes
Enter unit price: 2500
Enter quantity: 1

--- ORIGINAL SHOPPING BILL ---
Subtotal: Rs.6500.00
Discount: Rs.650.00
Final Amount: Rs.5850.00
Budget: Rs.5000.00

Budget exceeded by Rs.850.00

--- REVISED SHOPPING BASKET ---
Shoes: Quantity = 0, Cost = Rs.0.00
Rice: Quantity = 2, Cost = Rs.2400.00
Oil: Quantity = 2, Cost = Rs.1600.00
Revised Subtotal: Rs.4000.00
Discount: Rs.0.00
Final Amount: Rs.4000.00
Budget Remaining: Rs.1000.00
Units reduced: 1